HCIA-AI V4.0 — Recent Real Exam Recalled Questions & Verified Solutions

Authentic Test-Taker Exam Recall • 41 High-Yield Questions with Verified Answers, Math Derivations & Exam Traps

Curated by: **Umed Jamal Nouri** • **v1.1 (Nov 2026)** • **github.com/UmedJaMala**

RECENT EXAM SOURCE BREAKDOWN & PASSING ROADMAP

1. Direct Candidate Exam Experience
- These 41 questions were directly recalled by a successful candidate from the actual **HCIA-AI V4.0** computer-based examination.
- Passing Score: 600 / 1000 (typically 60 questions in 90 minutes).

2. High-Yield Question Types Tested
- **Tensor Shape Math:** (3, 2, 2) & (2, 2, 2) 3D array indexing.
- **Neuron Activation:** $\sum w_i x_i$ with sign(x) outputting **+1**.
- **PyTorch & MindSpore:** nn.Linear, CKPT, dynamic graphs.

3. Core Traps & Key Distinctions
- **Fraud vs Spam:** Fraud prioritizes **Recall**; Spam prioritizes **Precision**.
- **Foundation Models:** Large models do **NOT** completely replace small models!
- **FNNs vs RNNs:** FNN is strictly a DAG with **zero feedback loops**.

4. Framework Preprocessing Limits
- Frameworks (PyTorch/TensorFlow) provide tensor transformations, but **do not** provide raw business data collection and cleaning.
- BERT is **Encoder-only**; GPT is **Decoder-only**.

Complete Recalled Exam Questions & Detailed Verified Solutions (Q1 – Q41)

Question 1

SINGLE-ANSWER QUESTION • 16 Points

AI development relies on various factors. Which of the following enhances AI learning capabilities?

A. Computing Power (High-performance AI Processors & Clusters)
B. High-Quality Big Data
C. Advanced Deep Learning Algorithms
D. All of the above

CORRECT ANSWER: D (All of the above)

Detailed Technical Breakdown & Exam Trap: Per HCIA-AI V4.0 Chapter 1: The modern development of Artificial Intelligence is driven by the synergistic integration of three core cornerstones: **Data** (the foundational fuel), **Algorithms** (the computational models), and **Computing Power** (the hardware accelerator engine). Together, they enhance AI learning capabilities.

Question 2

TRUE OR FALSE • 16 Points

True or False: Foundation models outperform and completely replace small models in all application scenarios.

A. True
B. False

CORRECT ANSWER: B (False)

Detailed Technical Breakdown & Exam Trap: Exam Trap: While foundation models (large models) offer superior multi-task generalization, they do **NOT** completely replace small models. Small and lightweight models (e.g., MobileNet, TinyBERT) remain indispensable in edge computing, embedded IoT devices, ultra-low latency real-time tasks, and power-constrained environments where large models are too computationally expensive.

Question 3

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as common types of dirty data encountered during machine learning data cleaning?

A. Duplicate data records
B. Values with logical errors (inconsistent values)
C. Missing values (null or empty fields)
D. Values with formatting errors (data type mismatches)

CORRECT ANSWER: A, B, C, D

Detailed Technical Breakdown & Exam Trap: Per HCIA-AI V4.0 Chapter 2: In data preprocessing, dirty data encompasses four major anomalies: 1) duplicate records, 2) missing values, 3) logical/outlier errors (e.g. negative age), and 4) formatting errors (inconsistent dates, strings in numeric fields).

Question 4

TRUE OR FALSE • 16 Points

True or False: Machine learning is well suited for solving problems when the solution logic is complex or when the data volume is very large.

A. True

B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Machine learning is specifically advantageous where hand-crafting explicit rule-based algorithms is impractical due to high complexity, or where massive datasets contain hidden statistical patterns that algorithms can extract automatically.

Question 5

TRUE OR FALSE • 16 Points

True or False: Events or attributes that reflect the particular data or nature of a sample are called features.

A. True

B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Verbatim from HCIA-AI V4.0 terminology: A **feature** (or attribute) is an individual measurable property or characteristic of a sample that reflects its underlying nature and serves as the model's explanatory input variable.

Question 6

FILL-IN-THE-BLANK QUESTION • 16 Points

Fill in the blank: _____ Search performs an exhaustive search over all possible hyperparameter combinations to form a hyperparameter grid.

Answer: _____

CORRECT ANSWER: Grid (or Grid Search)

Detailed Technical Breakdown & Exam Trap: Grid Search (e.g. GridSearchCV in scikit-learn) evaluates every combination in a Cartesian product grid of hyperparameter values, ensuring an exhaustive exploration across discrete parameter spaces.

Question 7

TRUE OR FALSE • 16 Points

True or False: Classification and regression are two major types of prediction tasks; classification outputs discrete labels while regression outputs a continuous set.

A. True

B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Supervised machine learning prediction tasks fall into two fundamental categories: **Classification** predicts discrete categorical labels (e.g., spam/not spam, class 0/1/2), while **Regression** predicts continuous numeric values (e.g., stock price, temperature).

Question 8

TRUE OR FALSE • 16 Points

True or False: Data dimensionality reduction aims to reduce data volume and improve model accuracy and training efficiency.

A. True

B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Dimensionality reduction techniques (such as PCA) transform high-dimensional feature spaces into lower dimensions, mitigating the 'curse of dimensionality', removing multicollinear noise, and improving computational training efficiency.

Question 9

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a standard mathematical criterion used by decision tree algorithms to split nodes?

A. Information Gain (ID3 algorithm)
B. Gini Impurity Index (CART algorithm)
C. Gain Ratio (C4.5 algorithm)
D. Sigmoid Log-Loss Residual

CORRECT ANSWER: D (Sigmoid Log-Loss Residual)

Detailed Technical Breakdown & Exam Trap: Decision trees split nodes using Information Gain (ID3), Gain Ratio (C4.5), and Gini Index (CART). Sigmoid Log-Loss is an optimization loss function used in Logistic Regression and Neural Networks, not a decision tree splitting criterion.

Question 10

SINGLE-ANSWER QUESTION • 16 Points

The learning signal provided by the environment evaluates the quality of actions taken, and the learning system aims to maximize the cumulative reward signal by mapping from the environment state to actions. Which type of learning does this describe?

A. Supervised Learning
B. Unsupervised Learning
C. Reinforcement Learning
D. Semi-Supervised Learning

CORRECT ANSWER: C (Reinforcement Learning)

Detailed Technical Breakdown & Exam Trap: Reinforcement Learning (RL) operates via an agent interacting with an environment through states, actions, and scalar rewards, learning an optimal policy to maximize cumulative long-term reward.

Question 11

SINGLE-ANSWER QUESTION • 16 Points

Which of the following stages are mandatory when building a model using machine learning?

A. Data Preparation and Model Training
B. Direct deployment without offline validation
C. Transforming all tabular data into text tokens
D. Deploying exclusively on Ascend 910 clusters

CORRECT ANSWER: A (Data Preparation and Model Training)

Detailed Technical Breakdown & Exam Trap: In the standard machine learning workflow, **Data Preparation** (cleaning/preprocessing) and **Model Training** (optimizing parameters on training data) are mandatory core requirements.

Question 12

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about machine learning is FALSE?

A. Machine learning models require zero training data and learn entirely without prior examples
B. Supervised learning requires ground-truth labels for feature examples
C. Overfitting occurs when a model generalizes poorly to unseen test data
D. Hyperparameters are configured prior to model training

CORRECT ANSWER: A

Detailed Technical Breakdown & Exam Trap: Statement A is blatantly false. Machine learning is fundamentally empirical: it learns statistical patterns and decision boundaries directly from training data examples.

Question 13

SINGLE-ANSWER QUESTION • 16 Points

When deploying a machine learning model for credit card fraud detection and spam email filtering, which evaluation metric is more critical for each scenario respectively?

A. Fraud Detection: Recall; Spam Email Filtering: Precision
B. Fraud Detection: Precision; Spam Email Filtering: Recall
C. Fraud Detection: Accuracy; Spam Email Filtering: Accuracy
D. Fraud Detection: Specificity; Spam Email Filtering: Specificity

CORRECT ANSWER: A (Fraud: Recall; Spam: Precision)

Detailed Technical Breakdown & Exam Trap: Exam Classic: In Credit Card Fraud, missing a fraudulent transaction (False Negative) results in catastrophic financial loss, making **Recall** paramount. In Spam Filtering, misclassifying an important legitimate email as spam (False Positive) is unacceptable, making **Precision** paramount.

Question 14

TRUE OR FALSE • 16 Points

True or False: In the self-attention mechanism, the Query, Key, and Value vectors are generated through different linear transformations of the input sequence.

A. True
B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: In Self-Attention (Transformer architecture), the input token matrix X is multiplied by three distinct learnable projection weight matrices: $Q = X W_Q$, $K = X W_K$, and $V = X W_V$.

Question 15

SINGLE-ANSWER QUESTION • 16 Points

A feedforward neural network has one hidden layer with three neurons having output values of 10, -2, and 6. The weights connecting these neurons to the first neuron in the output layer are 0.1, 0.2, and 0.01, respectively (assume bias = 0). After passing through the sign(x) activation function, what is the output of the neuron?

A. -1
B. 0
C. +1
D. 0.66

CORRECT ANSWER: C (+1)

Detailed Technical Breakdown & Exam Trap: Step-by-Step Calculation:
1) Compute net weighted input sum: $z = \sum x_i w_i = (10 \times 0.1) + (-2 \times 0.2) + (6 \times 0.01)$
2) $z = 1.0 - 0.4 + 0.06 = 0.66$
3) The activation function is sign(x), where sign(z) = +1 for $z > 0$, -1 for $z < 0$, and 0 for $z = 0$.
Since $z = 0.66 > 0$, the activation output is exactly **+1**.

Question 16

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements about feedforward neural networks (FNNs) is FALSE?

A. Information propagates strictly unidirectionally from input to output layers
B. FNN network topologies can be represented as Directed Acyclic Graphs (DAGs)
C. Neurons form internal cyclic feedback connections to store temporal sequence memories
D. Neurons within the same layer do not connect to one another

CORRECT ANSWER: C

Detailed Technical Breakdown & Exam Trap: FNNs transmit information strictly forward without feedback cycles. Cyclic feedback connections belong to **Recurrent Neural Networks (RNNs)**, not FNNs.

Question 17

SINGLE-ANSWER QUESTION • 16 Points

Both Sigmoid and Tanh functions are traditional activation functions. Which statement comparing them is TRUE?

A. Tanh is zero-centered with output range (-1, 1), whereas Sigmoid has output range (0, 1)
B. Sigmoid prevents vanishing gradients in networks deeper than 10 layers
C. The maximum derivative of Sigmoid is 1.0 at x = 0
D. Tanh outputs strictly non-negative values

CORRECT ANSWER: A

Detailed Technical Breakdown & Exam Trap: Tanh maps inputs to $(-1, 1)$ and is zero-centered (which assists gradient descent optimization). Sigmoid maps to $(0, 1)$ and is not zero-centered. Both suffer from gradient saturation (Sigmoid max derivative is only 0.25).

Question 18

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements describe the recognized advantages of the ReLU (Rectified Linear Unit) activation function?

A. For positive inputs (x > 0), its derivative is constant (1), preventing vanishing gradients
B. It involves simple thresholding max(0, x), which computes substantially faster than exponential functions
C. It introduces sparsity by outputting exactly 0 for negative inputs
D. It completely prevents neurons from ever dying during training

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: ReLU $f(x) = \max(0, x)$ computes very fast, avoids vanishing gradients for positive activations ($f'(x) = 1$), and introduces sparsity. (D is false because ReLU can suffer from the 'Dying ReLU' problem when large gradients cause neurons to output 0 permanently).

Question 19

SINGLE-ANSWER QUESTION • 16 Points

Feedforward neural networks are among the most commonly used artificial neural networks. Which of the following statements is FALSE about FNNs?

A. An FNN consists of an input layer, one or more hidden layers, and an output layer
B. FNNs can store information across time steps through cyclical feedback loops
C. With non-linear activations, multi-layer FNNs can approximate continuous functions (Universal Approximation)
D. In an artificial neuron, the scalar bias term b acts as the activation threshold

CORRECT ANSWER: B

Detailed Technical Breakdown & Exam Trap: FNNs possess zero feedback loops and zero internal temporal state memory; sequence memory requires recurrent structures (RNN / LSTM).

Question 20

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques are effective methods to mitigate overfitting in deep neural networks?

A. Applying L1 or L2 weight regularization (weight decay)
B. Incorporating Dropout layers during training
C. Performing Data Augmentation to expand training sample diversity
D. Disabling validation checks and training for 100,000 extra epochs

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: Overfitting is mitigated by: 1) Regularization (L1/L2), 2) Dropout, 3) Data Augmentation, and 4) Early Stopping. Training endlessly without regularization worsens overfitting.

Question 21

TRUE OR FALSE • 16 Points

True or False: A convolutional layer in a CNN can share parameters to decrease the total number of network parameters; this sharing ensures that the convolution kernel parameters within the same layer are identical while differing across different layers.

A. True
B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Weight Sharing: In a CNN, a convolution filter slides across the entire receptive field using the exact same kernel weights, drastically reducing parameters compared to fully connected layers while learning shift-invariant features.

Question 22

TRUE OR FALSE • 16 Points

True or False: In Long Short-Term Memory (LSTM) networks, the forget gate decides what information to remove from the cell state; discarded information is assigned a coefficient of 0, and retained information is assigned a coefficient of 1.

A. True
B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: The LSTM forget gate $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$ uses a Sigmoid activation yielding values between 0 and 1: a coefficient of 0 completely wipes out past information, while 1 preserves it entirely.

Question 23

SINGLE-ANSWER QUESTION • 16 Points

Which of the following layers in a Convolutional Neural Network (CNN) work synergistically together to achieve spatial translation invariance in image recognition?

A. Convolutional Layer and Pooling Layer
B. Fully Connected Layer and Softmax Layer
C. Dropout Layer and Batch Normalization Layer
D. Flatten Layer and Dense Layer

CORRECT ANSWER: A (Convolutional Layer and Pooling Layer)

Detailed Technical Breakdown & Exam Trap: The combination of **Convolution** (local receptive fields and parameter sharing) and **Pooling** (subsampling/downsampling) enables CNNs to achieve spatial translation invariance.

Question 24

SINGLE-ANSWER QUESTION • 16 Points

In deep learning, a large number of stacked layers can lead to the vanishing gradient problem. In which of the following training phases does this issue typically manifest?

A. Forward Propagation
B. Backpropagation (Backward Pass)
C. Data Augmentation
D. Online Inference Deployment

CORRECT ANSWER: B (Backpropagation)

Detailed Technical Breakdown & Exam Trap: Vanishing gradients occur during **Backpropagation** when calculating gradients via the chain rule. Repeated multiplication of small derivative values (e.g. Sigmoid ≤ 0.25) across deep layers causes gradients to decay exponentially toward zero.

Question 25

MULTIPLE-ANSWER QUESTION • 18 Points

The Multi-Layer Perceptron (MLP) was developed to overcome the limitations of the single-layer perceptron. Which of the following statements are TRUE?

A. A single-layer perceptron can only form linear decision boundaries and strictly fails to solve XOR
B. An MLP incorporates one or more hidden layers equipped with non-linear activation functions
C. Stacking multiple dense layers without activation functions simply collapses into a single linear matrix multiplication
D. MLPs cannot be trained using gradient descent and backpropagation

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: Statements A, B, and C are true. (D is false because MLPs are trained using backpropagation and gradient descent).

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following foundation models is built using an encoder-only Transformer architecture?

A. GPT (Generative Pre-trained Transformer)
B. BERT (Bidirectional Encoder Representations from Transformers)
C. T5 (Text-to-Text Transfer Transformer)
D. LLaMA

CORRECT ANSWER: B (BERT)

Detailed Technical Breakdown & Exam Trap: Transformer Architecture Taxonomy:
• **Encoder-only:** BERT (for bidirectional text comprehension, classification, NER).
• **Decoder-only:** GPT, LLaMA, DeepSeek (for autoregressive text generation).
• **Encoder-Decoder:** T5, BART (for sequence-to-sequence translation and summarization).

Question 27

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods are effective strategies to address class imbalance in deep learning datasets?

A. Synthetic Minority Over-sampling Technique (SMOTE)
B. Random undersampling of the majority class
C. Applying Focal Loss or weighted cross-entropy loss penalties
D. Deleting all minority class samples from the dataset

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: Class imbalance is resolved by data-level techniques (SMOTE oversampling, undersampling) and cost-sensitive algorithm-level techniques (Focal Loss, class-weighted cross entropy).

Question 28

MULTIPLE-ANSWER QUESTION • 18 Points

Weight sharing is a fundamental design principle in deep learning. Which of the following neural network architectures utilize weight sharing?

A. Convolutional Neural Networks (CNN)
B. Recurrent Neural Networks (RNN)
C. Traditional Fully Connected Multi-Layer Perceptron (MLP)
D. Unweighted lookup tables

CORRECT ANSWER: A, B

Detailed Technical Breakdown & Exam Trap: CNNs share convolution kernel weights across spatial positions of the image; RNNs share weight matrices across sequential time steps. Traditional MLPs use distinct independent weights for every connection.

Question 29

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding L1 and L2 regularization are TRUE?

A. L1 regularization (Lasso) drives non-essential parameter weights to exactly zero, creating sparse models
B. L2 regularization (Ridge / Weight Decay) penalizes the squared sum of weights, keeping weights small
C. Both L1 and L2 regularization help prevent model overfitting
D. L1 regularization always produces denser models than L2 regularization

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: Statements A, B, and C are correct. L1 promotes parameter sparsity (feature selection), while L2 shrinks weights smoothly toward zero.

Question 30

TRUE OR FALSE • 16 Points

True or False: Dropout randomly disables some neurons during training, whereby the parameters of these disabled neurons will not participate in the next round of training.

A. True
B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: During training, Dropout randomly zeros out a proportion p of hidden unit activations. The dropped neurons do not contribute to forward propagation and receive no parameter updates during backpropagation in that mini-batch iteration.

Question 31

MULTIPLE-ANSWER QUESTION • 18 Points

In deep learning, data augmentation is an effective method for preventing overfitting. Which of the following describe valid image data augmentation techniques?

A. Random horizontal flipping, cropping, and rotation
B. Adjusting image brightness, contrast, and saturation (color jitter)
C. Adding random Gaussian noise to pixel values
D. Altering ground-truth target labels at random

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: Valid data augmentation applies label-preserving geometric transformations (flipping, cropping, rotating) and photometric transformations (color jitter, noise). Target labels must remain accurate.

Question 32

SINGLE-ANSWER QUESTION • 16 Points

During model training, John saves the trained model in CKPT format. Which of the following statements about CKPT is FALSE?

A. CKPT is a widely used checkpoint serialization format in Huawei MindSpore and TensorFlow
B. The CKPT format uses Google Protocol Buffers to serialize weights and optimizer states
C. CKPT files can only be used for offline inference and cannot be used to resume interrupted training
D. Saving checkpoints allows models to be restored in case of hardware interruption

CORRECT ANSWER: C

Detailed Technical Breakdown & Exam Trap: Statement C is strictly FALSE. The primary purpose of .ckpt files is precisely to store weights and optimizer states so that interrupted training can be seamlessly resumed and fine-tuned.

Question 33

SINGLE-ANSWER QUESTION • 16 Points

What is the dimensional shape of the nested tensor [[[1, 2], [3, 4]], [[5, 6], [7, 8]], [[9, 10], [11, 12]]]?

A. (3, 2, 2)
B. (2, 3, 2)
C. (2, 2, 3)
D. (12,)

CORRECT ANSWER: A ((3, 2, 2))

Detailed Technical Breakdown & Exam Trap: Tensor Shape Derivation:
• Axis 0 (Outer list): contains 3 sub-blocks $\rightarrow$ size **3**.
• Axis 1 (Middle list): each block contains 2 rows $\rightarrow$ size **2**.
• Axis 2 (Inner list): each row contains 2 scalars $\rightarrow$ size **2**.
Hence, the shape is (3, 2, 2) with $3 \times 2 \times 2 = 12$ total elements.

Question 34

MULTIPLE-ANSWER QUESTION • 18 Points

John wants to use PyTorch to implement a deep learning model for machine translation. Which of the following statements about his implementation operations are TRUE?

A. He can design an Encoder-Decoder architecture with Attention mechanism (or Transformer)
B. He should define model layers and forward computation by subclassing torch.nn.Module
C. He must manually derive and write mathematical gradient update formulas without using autograd
D. He can use CrossEntropyLoss to compute token classification loss across the target vocabulary

CORRECT ANSWER: A, B, D

Detailed Technical Breakdown & Exam Trap: Statements A, B, and D are standard deep learning practice. C is false because PyTorch's automatic differentiation engine (loss.backward()) automatically computes gradients.

Question 35

SINGLE-ANSWER QUESTION • 16 Points

What is the dimensional shape of the nested tensor [[[0, 1], [2, 3]], [[4, 5], [6, 7]]]?

A. (2, 2, 2)
B. (2, 4)
C. (4, 2)
D. (8,)

CORRECT ANSWER: A ((2, 2, 2))

Detailed Technical Breakdown & Exam Trap: Tensor Shape Derivation:
• Axis 0: contains 2 outer blocks $\rightarrow$ size **2**.
• Axis 1: each block contains 2 rows $\rightarrow$ size **2**.
• Axis 2: each row contains 2 numbers $\rightarrow$ size **2**.
Hence, the shape is (2, 2, 2) with $2 \times 2 \times 2 = 8$ total elements.

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is a primary core characteristic of the PyTorch deep learning framework?

A. Dynamic computation graphs (Define-by-Run) and eager execution
B. Pure static-only graph execution with no interactive debugging
C. Requiring separate C++ compilation for every basic tensor addition
D. Inability to execute on GPU or NPU hardware accelerators

CORRECT ANSWER: A (Dynamic computation graphs and eager execution)

Detailed Technical Breakdown & Exam Trap: PyTorch's defining characteristic is dynamic graph computation (Define-by-Run), executing operations immediately in eager mode, which provides intuitive debugging and seamless integration with standard Python.

Question 37

TRUE OR FALSE • 16 Points

True or False: Current mainstream AI development frameworks (such as PyTorch and MindSpore) support compatibility between dynamic computation graphs and static computation graphs.

A. True
B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Modern AI frameworks support both paradigms: PyTorch offers eager dynamic execution alongside TorchScript and torch.compile for static graphs; MindSpore natively supports both ms.PYNATIVE_MODE (dynamic) and ms.GRAPH_MODE (static).

Question 38

TRUE OR FALSE • 16 Points

True or False: AI development frameworks such as PyTorch and TensorFlow provide complete built-in preprocessing functions for raw business data collection, data cleaning, feature selection, and normalization.

A. True
B. False

CORRECT ANSWER: B (False)

Detailed Technical Breakdown & Exam Trap: Exam Trap: While deep learning frameworks provide tensor transformation utilities (e.g. torchvision.transforms for resizing and normalization), they do **NOT** provide end-to-end built-in systems for raw business data collection, database cleaning, and feature selection. Those tasks are handled by dedicated tools (e.g. Pandas, Scikit-Learn, Spark, ModelArts).

Question 39

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI development frameworks can John use to build and train a deep learning model for speech recognition?

A. PyTorch (with torchaudio)
B. Huawei MindSpore (with mindspore.dataset.audio)
C. TensorFlow (with tf.audio)
D. Standard Linux Bash shell without programming libraries

CORRECT ANSWER: A, B, C

Detailed Technical Breakdown & Exam Trap: PyTorch, MindSpore, and TensorFlow all provide comprehensive audio processing and neural network modeling libraries for automatic speech recognition (ASR).

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following PyTorch APIs is used to create a fully connected (dense) linear transformation layer?

A. torch.nn.Linear
B. torch.nn.Dense
C. torch.nn.Conv2d
D. torch.nn.FullyConnected

CORRECT ANSWER: A (torch.nn.Linear)

Detailed Technical Breakdown & Exam Trap: In PyTorch, fully connected layers are instantiated as torch.nn.Linear(in_features, out_features). (In MindSpore, the equivalent is nn.Dense).

Question 41

TRUE OR FALSE • 16 Points

True or False: John needs to determine the hyperparameters for training when designing a deep learning network model; hyperparameters must be fixed during training and cannot be changed randomly.

A. True

B. False

CORRECT ANSWER: A (True)

Detailed Technical Breakdown & Exam Trap: Hyperparameters (learning rate, batch size, number of hidden layers, optimizer selection) are external configuration settings established by the engineer before training begins; they govern the training dynamics and are not learned by gradient descent during the training run.

HCIA-AI V4.0 Recalled Exam: Master Quick-Review Answer Key (All 41 Questions)

Q#	Question Type	Tested Knowledge Concept	Verified Correct Answer
Q1	Single-Answer Question	AI development relies on various factors. Which of the ...	D (All of the above)
Q2	True or False	True or False: Foundation models outperform and complet...	B (False)
Q3	Multiple-Answer Question	Which of the following are recognized as common types o...	A, B, C, D
Q4	True or False	True or False: Machine learning is well suited for solv...	A (True)
Q5	True or False	True or False: Events or attributes that reflect the pa...	A (True)
Q6	Fill-in-the-Blank Question	Fill in the blank: _____ Search performs an exhaustive...	Grid (or Grid Search)
Q7	True or False	True or False: Classification and regression are two ma...	A (True)
Q8	True or False	True or False: Data dimensionality reduction aims to re...	A (True)
Q9	Single-Answer Question	Which of the following is NOT a standard mathematical c...	D (Sigmoid Log-Loss Residual)
Q10	Single-Answer Question	The learning signal provided by the environment evaluat...	C (Reinforcement Learning)
Q11	Single-Answer Question	Which of the following stages are mandatory when buildi...	A (Data Preparation and Model Training)
Q12	Single-Answer Question	Which of the following statements about machine learnin...	A
Q13	Single-Answer Question	When deploying a machine learning model for credit card...	A (Fraud: Recall; Spam: Precision)
Q14	True or False	True or False: In the self-attention mechanism, the Que...	A (True)
Q15	Single-Answer Question	A feedforward neural network has one hidden layer with ...	C (+1)
Q16	Single-Answer Question	Which of the following statements about feedforward neu...	C
Q17	Single-Answer Question	Both Sigmoid and Tanh functions are traditional activat...	A
Q18	Multiple-Answer Question	Which of the following statements describe the recogniz...	A, B, C
Q19	Single-Answer Question	Feedforward neural networks are among the most commonly...	B
Q20	Multiple-Answer Question	Which of the following techniques are effective methods...	A, B, C
Q21	True or False	True or False: A convolutional layer in a CNN can share...	A (True)
Q22	True or False	True or False: In Long Short-Term Memory (LSTM) network...	A (True)
Q23	Single-Answer Question	Which of the following layers in a Convolutional Neural...	A (Convolutional Layer and Pooling Layer)
Q24	Single-Answer Question	In deep learning, a large number of stacked layers can ...	B (Backpropagation)
Q25	Multiple-Answer Question	The Multi-Layer Perceptron (MLP) was developed to overc...	A, B, C
Q26	Single-Answer Question	Which of the following foundation models is built using...	B (BERT)
Q27	Multiple-Answer Question	Which of the following methods are effective strategies...	A, B, C
Q28	Multiple-Answer Question	Weight sharing is a fundamental design principle in dee...	A, B
Q29	Multiple-Answer Question	Which of the following statements regarding L1 and L2 r...	A, B, C
Q30	True or False	True or False: Dropout randomly disables some neurons d...	A (True)
Q31	Multiple-Answer Question	In deep learning, data augmentation is an effective met...	A, B, C
Q32	Single-Answer Question	During model training, John saves the trained model in ...	C
Q33	Single-Answer Question	What is the dimensional shape of the nested tensor [[[1...	A ((3, 2, 2))
Q34	Multiple-Answer Question	John wants to use PyTorch to implement a deep learning ...	A, B, D
Q35	Single-Answer Question	What is the dimensional shape of the nested tensor [[[0...	A ((2, 2, 2))
Q36	Single-Answer Question	Which of the following is a primary core characteristic...	A (Dynamic computation graphs and eager execution)
Q37	True or False	True or False: Current mainstream AI development framew...	A (True)
Q38	True or False	True or False: AI development frameworks such as PyTorc...	B (False)
Q39	Multiple-Answer Question	Which of the following AI development frameworks can Jo...	A, B, C
Q40	Single-Answer Question	Which of the following PyTorch APIs is used to create a...	A (torch.nn.Linear)
Q41	True or False	True or False: John needs to determine the hyperparamet...	A (True)